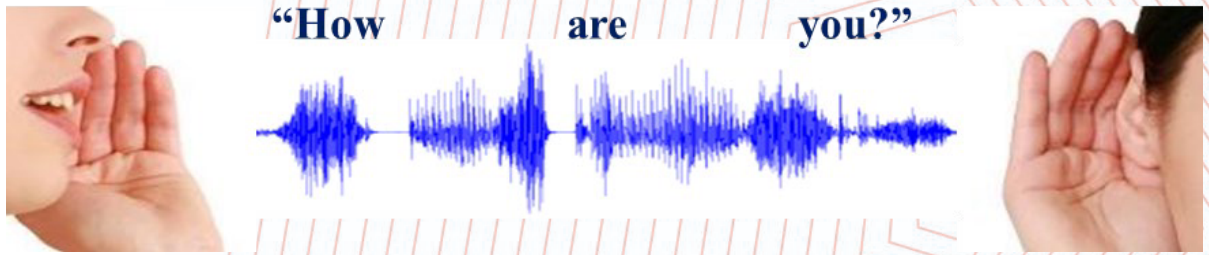
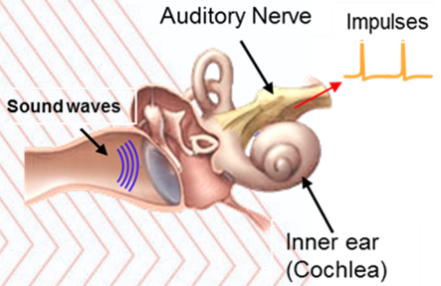
ELEC3104: Mini-Project – Cochlear Signal Processing





Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

ELEC3104: Mini-Project – Cochlear Signal Processing

Mini Project: Pass-Level: Implementation of a cascaded filter bank model of the cochlea

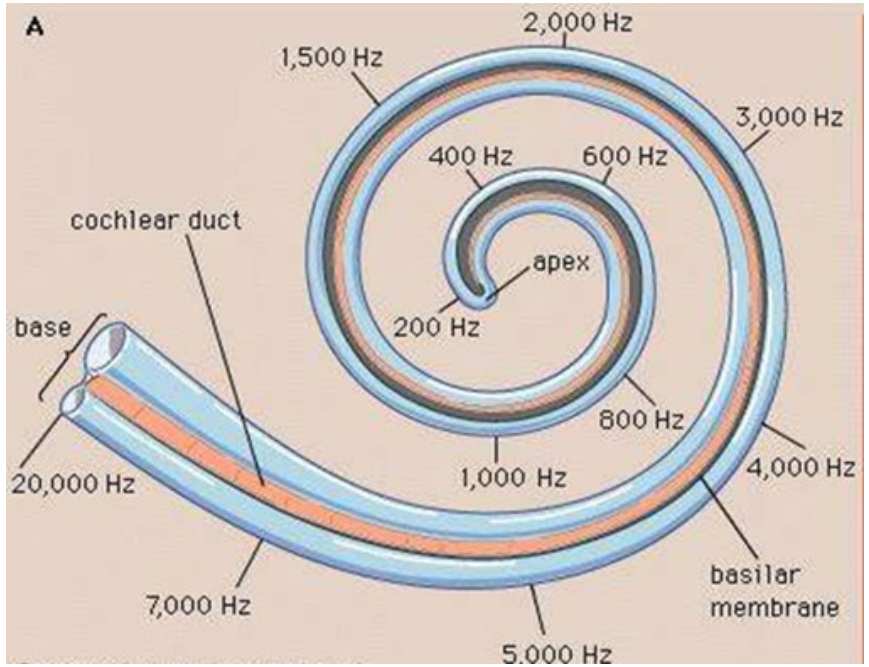
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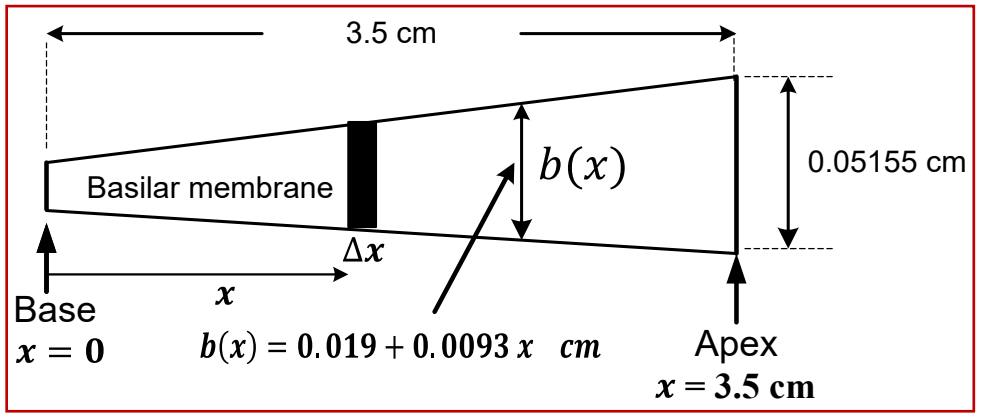
Pass-Level: Project Implementation

- Implement a digital transmission line model of the cochlea using 512 cascaded filter sections, each combining a resonator and a notch filter, at a sampling frequency of 16 kHz.
- Calculate the resonant frequencies, notch frequencies, quality factors, bandwidths and digital filter coefficients using the specified design assumptions.
- Obtain the pressure and displacement outputs at each section and implement one and two stages of spatial differentiation.
- Validate the model using an impulse, a single sinusoid and a sum of three sinusoids. Examine the impulse responses, magnitude responses and displacement patterns across sections.
- Demonstrate how different input frequencies produce maximum responses at different positions along the basilar membrane.
- Explain the roles of the resonators, notch filters and spatial differentiation, using evidence from your results.
- Present clearly labelled plots and reflect on the accuracy, limitations and behaviour of your implementation.

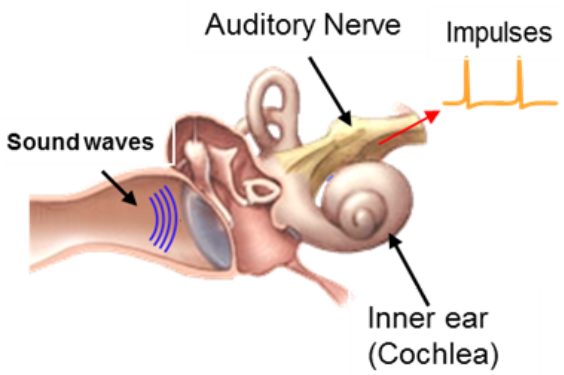
Inner ear (Cochlea)

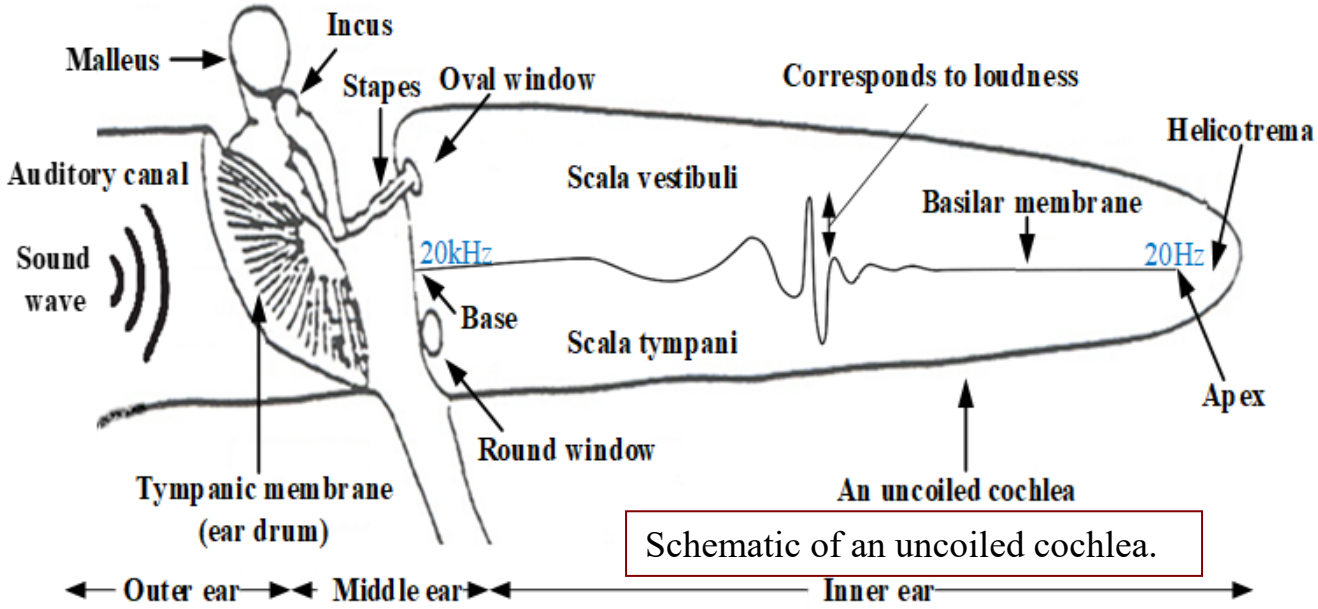


The cochlea performs frequency analysis: high frequencies produce maximum vibration near the base, while low frequencies peak towards the apex.



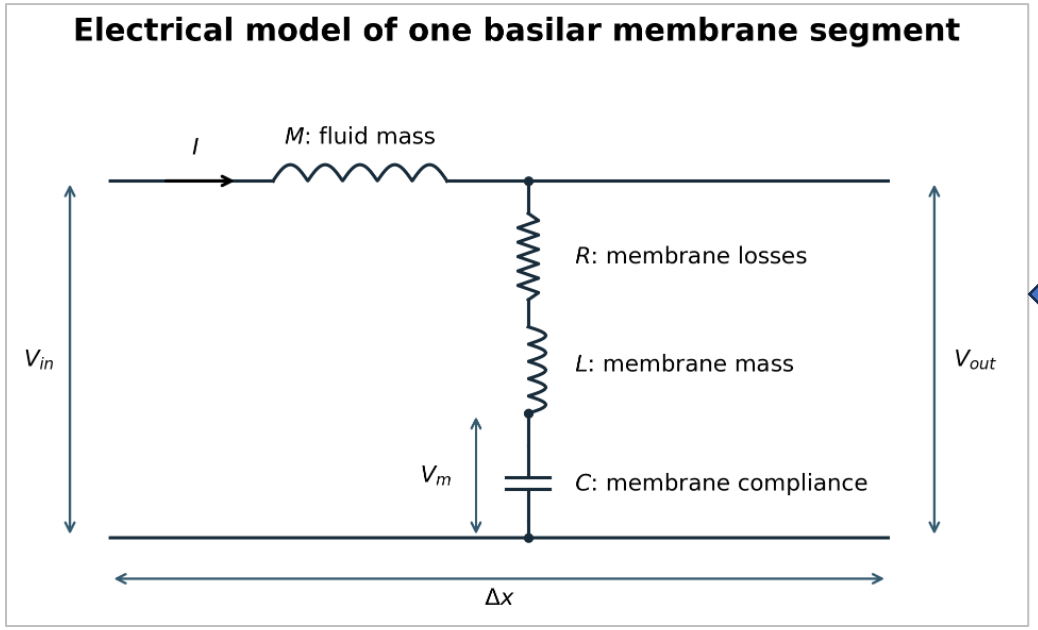
For modelling, the cochlea is represented in an uncoiled form, with position x measured along the basilar membrane from the base.





Schematic of an uncoiled cochlea.

Pass-Level: Project Implementation - Cochlear Modelling



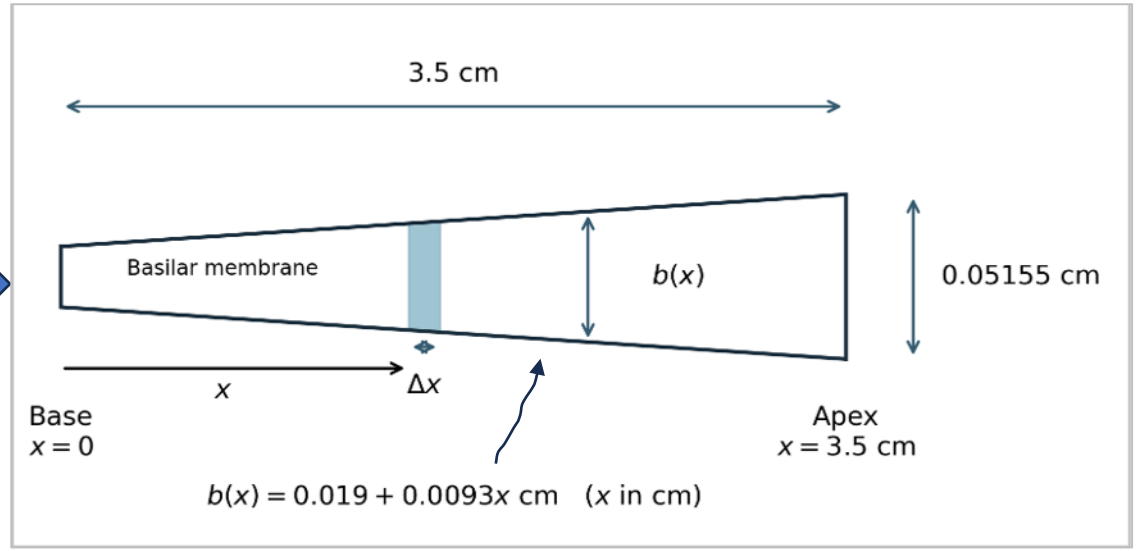
A small segment of the basilar membrane, of length Δx is represented by an electrical circuit. The series inductance M represents fluid mass, while R, L and C represent membrane losses, mass and compliance respectively. The voltage V_m is measured across the capacitor.

In this geometrical approximation, the basilar membrane is 3.5 cm long and increases in width from the base to the apex. Its width at position x is given by

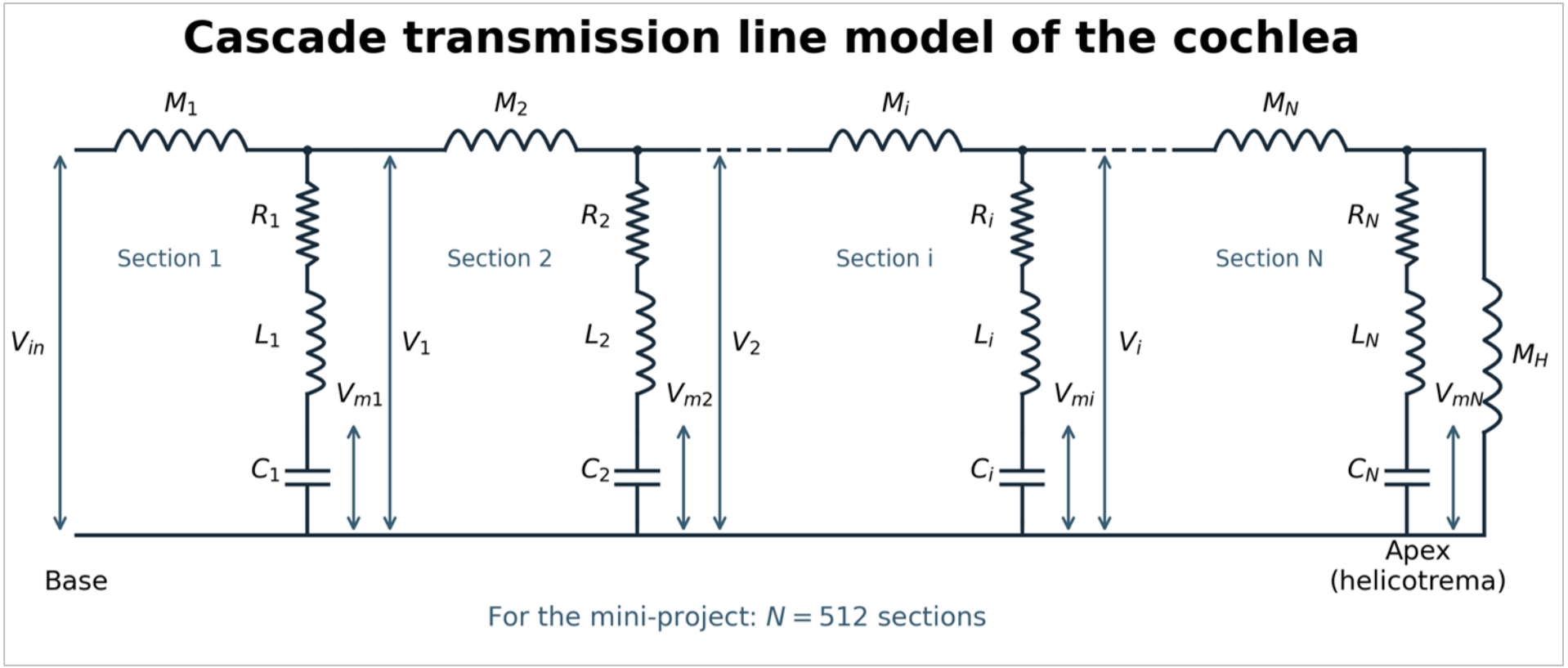
$$b(x) = 0.019 + 0.0093\, x \;\; cm$$

where x is measured in centimetres from the base. The shaded region represents one segment of length Δx .

Basilar Membrane in the Uncoiled Cochlea



Pass-Level: Project Implementation - Cochlear Modelling



The complete model is formed by connecting N electrical sections in cascade, from the base to the apex. Each section represents a local segment of the basilar membrane, with component values that vary along its length. The terminal inductance M_H represents the fluid mass associated with the helicotrema. In this mini-project, we will implement a digital version using N=512 sections.

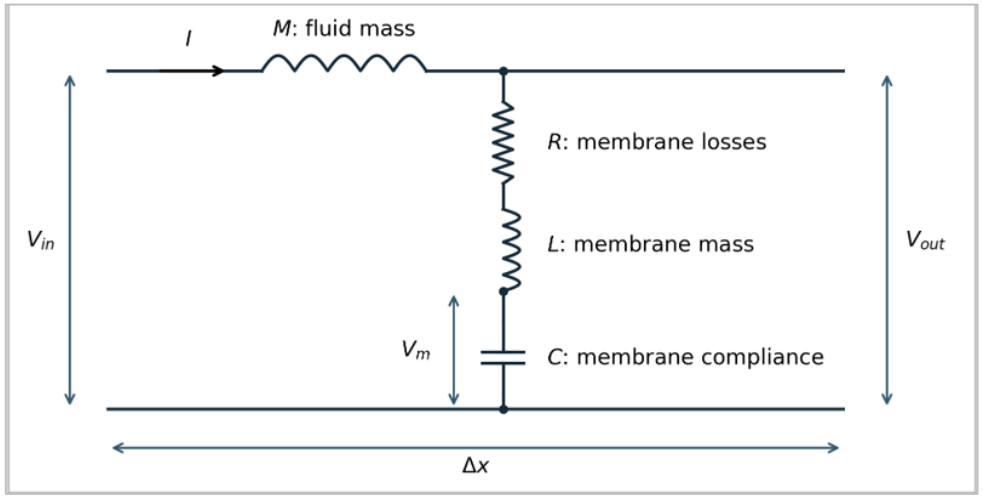
Pressure and Displacement Transfer Functions

✓ The Voltage (**pressure) transfer function** of an isolated section is:

$$\frac{V_o(s)}{V_i(s)} = \frac{\omega_p^2}{s^2 + B_p s + \omega_p^2} \cdot \frac{s^2 + B_z s + \omega_z^2}{\omega_z^2}$$

Resonator

Inverse Resonator or Notch Filter



✓ The **displacement transfer function** is:

$$\frac{V_m(s)}{V_i(s)} = \frac{\omega_p^2}{s^2 + B_p s + \omega_p^2}$$

✓ The displacement transfer function forms part of the pressure transfer function, allowing a simple cascade implementation.

ω_p: Pole angular/resonant frequency (rad/sec); ω_z: zero angular frequency (rad/sec)

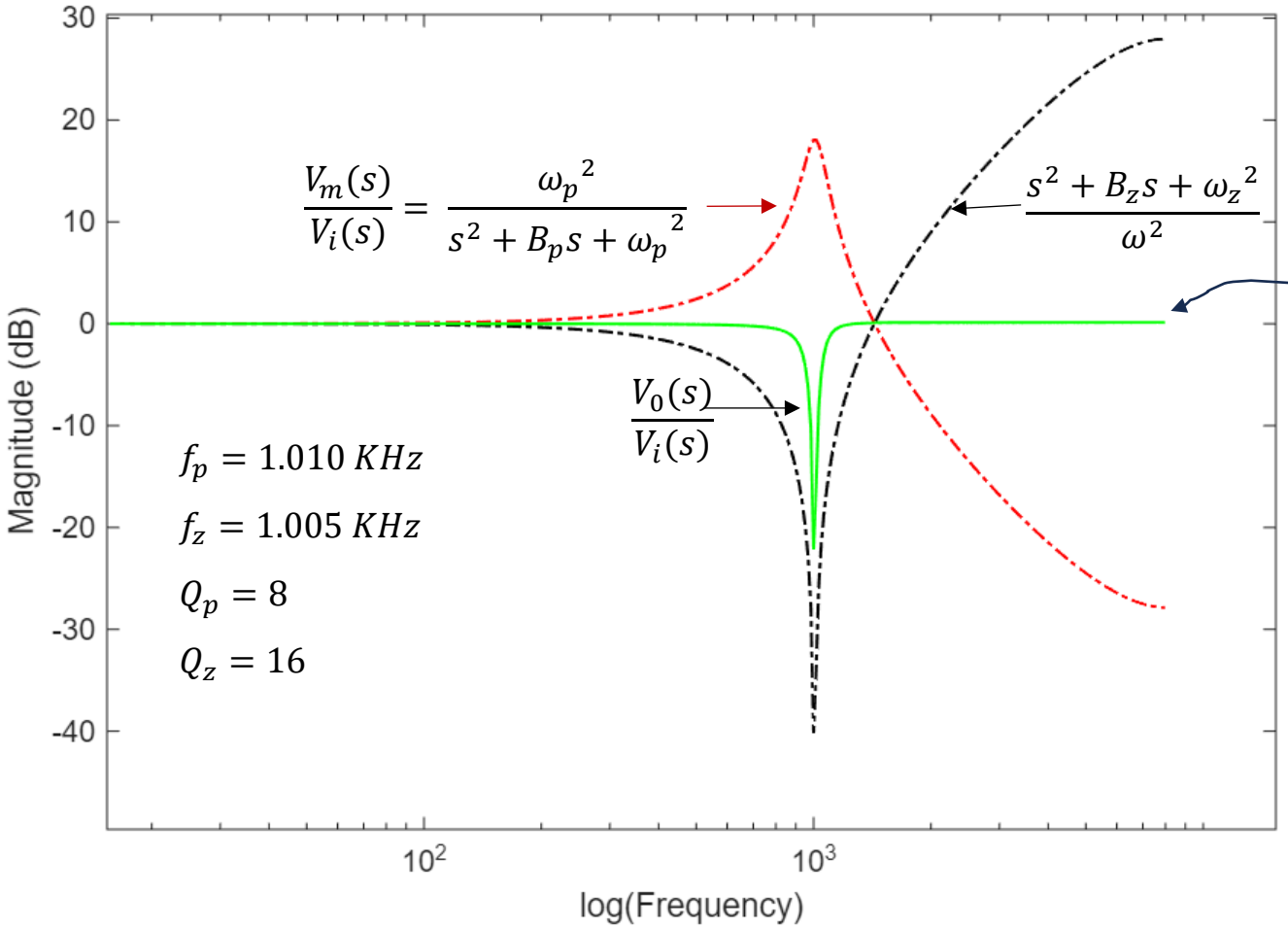
B_p: Pole bandwidth parameter (rad/sec); B_z: zero bandwidth parameter (rad/sec)

$V_i(s)$: Laplace transform of the input voltage, representing input pressure.

$V_o(s)$: Laplace transform of the output voltage, representing output pressure.

$V_m(s)$: Laplace transform of the capacitor voltage, representing membrane displacement.

Frequency Response - One Section of the Membrane



Pressure Transfer Function

$$\frac{V_{o(s)}}{V_{i(s)}} = \frac{{\omega_p}^2}{s^2+B_p s+{\omega_p}^2} \cdot \frac{s^2+B_z s+{\omega_z}^2}{{\omega_z}^2}$$

$Q_p = \frac{\omega_p}{B_p}$ and $Q_z = \frac{\omega_z}{B_z}$

Digital Filter Model of the Basilar Membrane (BM)

✓ A digital filter model of the basilar membrane can be obtained by pole-zero placement

$$\frac{V_o(s)}{V_i(s)} = \frac{\omega_p^2}{s^2 + B_p s + \omega_p^2} \cdot \frac{s^2 + B_z s + \omega_z^2}{\omega_z^2}$$

$$\frac{V_o(z)}{V_i(z)} = \frac{1 - b_1 + b_2}{1 - b_1 z^{-1} + b_2 z^{-2}} \cdot \frac{1 - a_1 z^{-1} + a_2 z^{-2}}{1 - a_1 + a_2}$$

Pressure Transfer Function

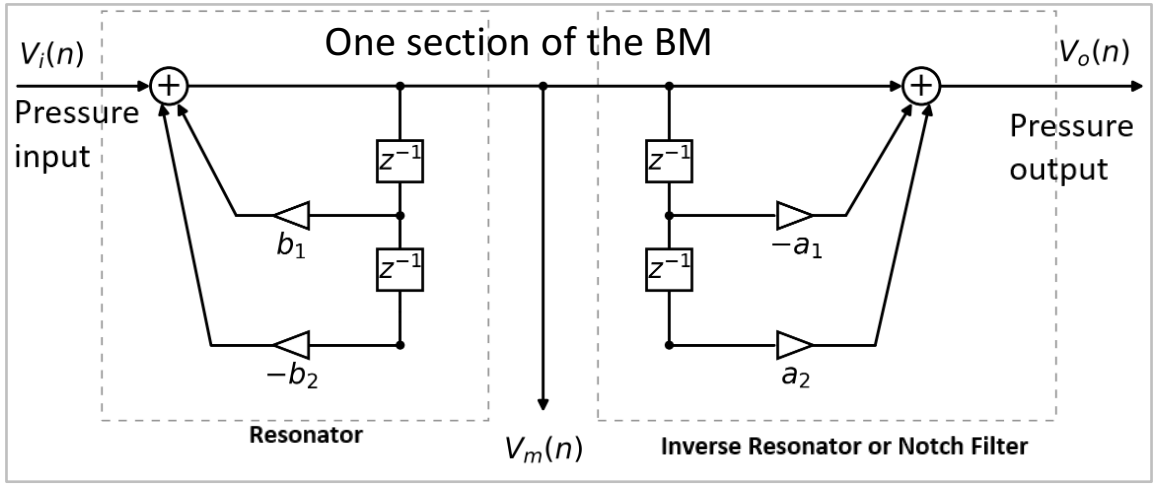
$$\frac{V_m(s)}{V_i(s)} = \frac{\omega_p^2}{s^2 + B_p s + \omega_p^2}$$

$$\frac{V_m(z)}{V_i(z)} = \frac{(1 - b_1 + b_2)}{1 - b_1 z^{-1} + b_2 z^{-2}}$$

Displacement Transfer Function

✓ Where $a_0, a_1, a_2, b_1, and\ b_2$ are the digital filter coefficients;

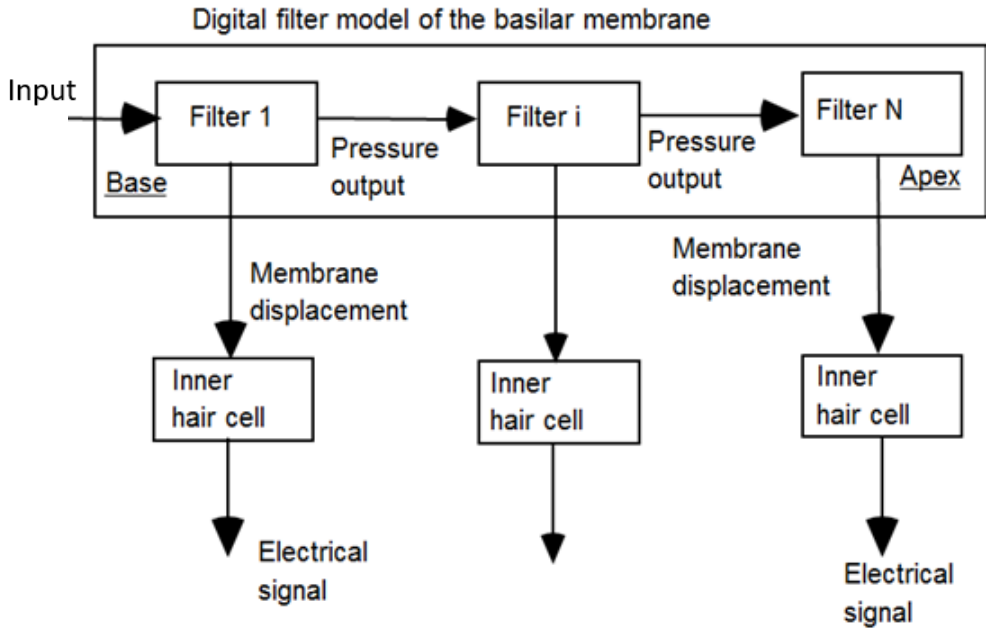
One can see that the displacement transfer function is contained in the pressure transfer function and hence a simple cascade arrangement is sufficient to represent the cochlear model



In each section, pressure is converted into displacement of the basilar membrane and transmitted to the following section. This leads to two transfer functions, one relating the output pressure, $V_o(n)$, and the input pressure, $V_i(n)$, and the other relating the output displacement, $V_m(n)$ to the input pressure.

Transmission Line Model of the Cochlea

- ✓ The cochlea is represented by a transmission line model comprising 512 cascaded digital filter sections (N = 512), incorporating low-pass filtering, notch filtering and resonance. The sampling frequency is 16 kHz.

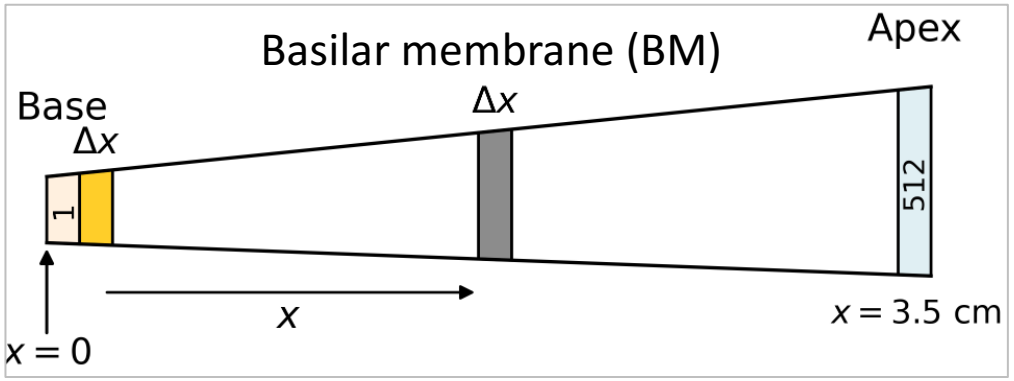


Transmission Line Model

- ✓ Each digital filter section represents a local region of the basilar membrane tuned to a specific frequency. Together, the 512 sections represent the entire basilar membrane.
- ✓ The input signal propagates through the cascaded sections, modelling the travelling pressure wave in the cochlear fluid.

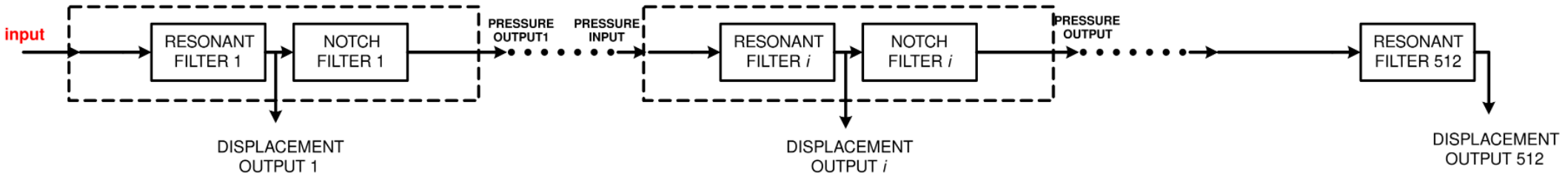
Selection of the Frequency Scale

- ✓ The model assumes a basilar membrane (BM) length of 3.5 cm, represented by 512 digital filter sections connected in cascade. Each section has the same length : Δx = 3.5/512 = 0.00684 cm;
- ✓ **Frequency scale:** Equal distances along the BM correspond to equal ratios of resonant frequency. The resonant frequency decreases exponentially from 7500 Hz at the base to approximately 35 Hz near the apex.



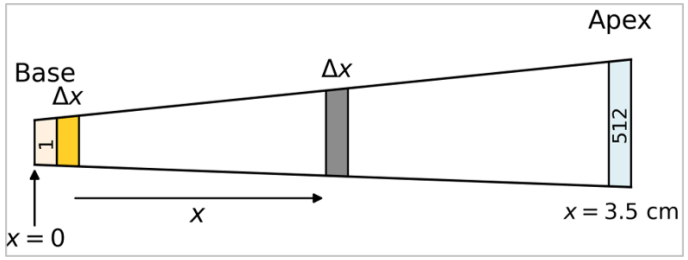
$$f_p(x_i) = 7500.00 \;\; 10^{-0.667 x_i} \; Hz \quad 0 \le x_i \le 3.5 \; cm$$

Filter number	Distance x_i in cm $x_i = (i-1)\Delta x$, $i = 1,2,\ldots,512.$	Resonant frequency, $f_p(x_i)$
1	**0**	**7500 Hz**
100	0.6768	2652.55 Hz
250	1.7021	549.20 Hz
512	**3.4932**	**35.08 Hz**



Each cascaded section represents one BM segment. Its displacement output is taken after the resonant filter, while the notch filter output provides the pressure input to the next section.

Implementation – Step 1: Calculate the Filter Frequencies



- ✓ Prepare a table containing the position, resonant frequency and notch frequency for all 512 filter sections.

- ✓ Number of filters N = 512; Length of the BM = 3.5 cm. $\Delta x = \frac{3.5}{512} = 0.00684\ cm$
- ✓ $x = 0, \Delta x, 2\Delta x, 3\Delta x, \ldots\ldots\ldots, 511\Delta x$; $\boldsymbol{Resonant\ frequency}$: $\boldsymbol{f_p(i) = (7500)\ 10^{-0.667\,(i-1)\Delta x}}$

Filter No (i)	Distance (x) cm	$f_p(i)$: Resonant Frequency (Hz)	$f_z(i)$: Notch Filter (Hz)
1	0	$f_p(1)$ = 7500	$f_z(1)$ = 7580 (Use 7580 Hz for the first section)
2	Δx	$f_p(2)$ = 7422	$f_z(2) = 0.985 \times f_p(1)$ = 7388
3	$2\Delta x$	$f_p(3)$ = 7344	$f_z(3) = 0.985 \times f_p(2)$ = 7310
4	$3\Delta x$	$f_p(4)$ = 7267	$f_z(4) = 0.985 \times f_p(3)$ = 7234
.		.	.
512	$511\Delta x$	$f_p(512)$ = 35.08	$f_z(512) = 0.985 \times f_p(511)$=34.92

- ✓ Set the notch frequency to 0.985 times the resonant frequency of the preceding section:

$$f_z(i) = 0.985\, f_p(i-1),$$

$$i = 2, \ldots, 512.$$

- ✓ This positions the notch close to the preceding section's resonance to reduce its accumulated resonant peak.

Implementation – Step 2: Quality Factors and Bandwidths

- ✓ For each of the 512 sections, use the resonant and notch frequencies calculated in Step 1.
- ✓ Set Q_p= 8 (Constant Q for all 512 filters)) and Q_z = 2 Q_p $= 16$, then calculate the corresponding bandwidths

$$BW_p(i) = \frac{f_p(i)}{Q_p(i)} \text{ and } BW_z(i) = \frac{f_z(i)}{Q_z(i)}$$

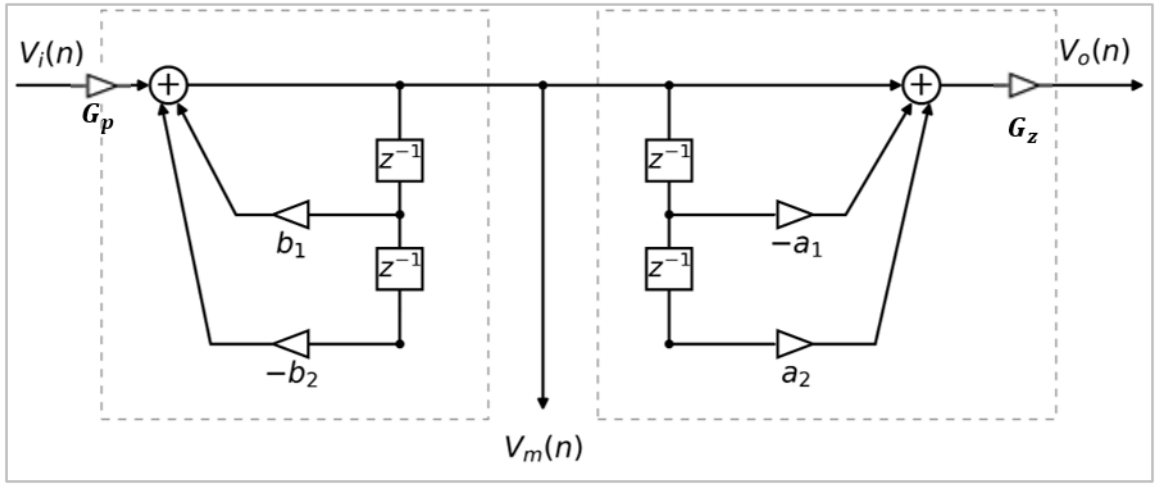
- Complete the table for all sections. With constant quality factors, the bandwidths decrease as the frequencies decrease towards the apex.
- Investigate how changing the quality factors affects the filter responses, while maintaining $Q_z >$ Q_p . A higher quality factor gives a narrower bandwidth.

Filter No (i):	$f_p(i)$ in Hz	$Q_p(i)$	$BW_p(i)$ in Hz	$f_z(i)$ in Hz	$Q_z(i)$	$BW_z(i)$ in Hz
1	7500	8	937.5	7580	16	473.75
2	7422	8	927.71	7388	16	461.72
3	7344	8	918.02	7310	16	456.90
.	.	.		.	.	
512	35.08	8	4.39	34.92	16	2.18

Design Criteria

- ✓ Use 512 cascaded sections with equally spaced positions and exponentially decreasing resonant frequencies.
- ✓ Select the notch frequencies using $f_z(i) = 0.985\ f_p(i-1)$ for i = 2,3, …. 512 with $f_z(1) = 7580\ Hz$
- ✓ Use constant quality factors $Q_p = 8\ and\ Q_z = 16$. Calculate each bandwidth using $BW = f/Q$
- ✓ Use pole-zero placement (Topic 4) to calculate the digital filter coefficients at $f_s = 16\ kHz$
- ✓ Ensure all poles lie strictly inside the unit circle and include the required gain normalisation.
- ✓ Verify the individual-section and overall cascade responses.

Implementation – Step 3: Filter Coefficients and Gain Normalisation



$$\frac{V_o(z)}{V_i(z)} = \frac{1 - b_1 + b_2}{1 - b_1 z^{-1} + b_2 z^{-2}} \cdot \frac{1 - a_1 z^{-1} + a_2 z^{-2}}{1 - a_1 + a_2}$$

$$\frac{V_m(z)}{V_i(z)} = \frac{(1 - b_1 + b_2)}{1 - b_1 z^{-1} + b_2 z^{-2}}$$

$$G_p = 1 - b_1 + b_2$$

$$G_z = \frac{1}{1 - a_1 + a_2}$$

Filter Number	f_p (Hz)	f_z (Hz)	Q_p	Q_z	BW_p(Hz)	BW_z (Hz)
1	7500	7580	8	16	937.5	473.75
2	7422	7388	8	16	927.71	461.72
3	7344	7310	8	16	918.02	456.90
.	.	.	.	.		
512	35.08	34.92	8	16	4.39	2.18

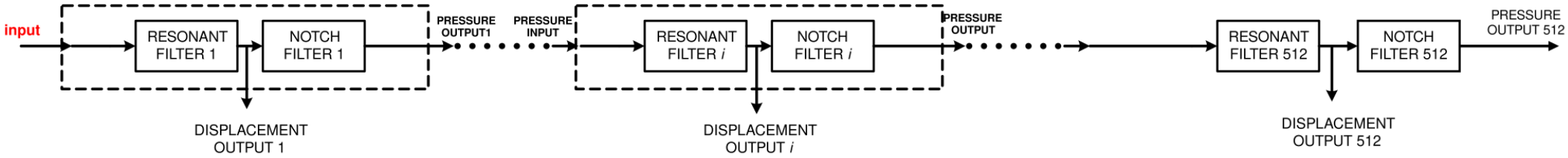
If your implementation is correct and you use the same assumptions, you should obtain the following parameter values for the selected filters

Filter Number	G_p	b_1	b_2	G_z	a_1 (Q_z = inf)	a_2 (Q_z = inf)
1	3.2662	-1.6005	0.6657	0.2517	-1.9729	1
2	3.2626	-1.5937	0.6689	0.2537	-1.9424	1
3	3.2574	-1.5854	0.6720	0.2546	-1.9271	1
.	.	.	.	.	.	.
.	.	.	.	.	.	.
512	0.0002	1.9981	0.9983	5316	1.9998	1

a_1' (Q_z = 16)	a_2' (Q_z= 16)
-1.7893	0. 8226
-1.7663	0.8269
-1.7542	0.8286
.	.
.	.
1.999	0.9991

Implementation – Step 4: Time-Domain Simulation

- Implement the cascaded model using MATLAB's filter() function. Use a sampling frequency of **16 kHz** and test the following inputs:
  - A unit impulse.
  - A single sinusoid: start with **1 kHz**, then investigate other frequencies.
- For each input, record the displacement and pressure outputs in two separate matrices. Each matrix should have **512 columns**, representing the sections, and one row per input sample.
- Apply the FFT to the recorded impulse responses to obtain the magnitude responses from the model input to each output. Use a sufficiently long impulse response to capture its decay, particularly for sections near the apex.
- For a sinusoidal input, plot a selected row of the displacement matrix against position along the BM. This shows the displacement across all sections at a particular instant. Allow the initial transient to decay before selecting the row.
- If spatial differentiation is required, calculate it across adjacent columns within each row, using the section spacing Δx.



Spatial Differentiation

Spatial Differentiation

- ✓ Spatial differentiation of the basilar membrane displacement is used in this model to approximate fluid-mediated coupling between the stereocilia of the inner hair cells.
- ✓ Let $s_m[n]$ denote the displacement output of section m at time sample n .
- ✓ The first spatial difference is calculated by subtracting the displacement outputs of adjacent sections at the same time sample:

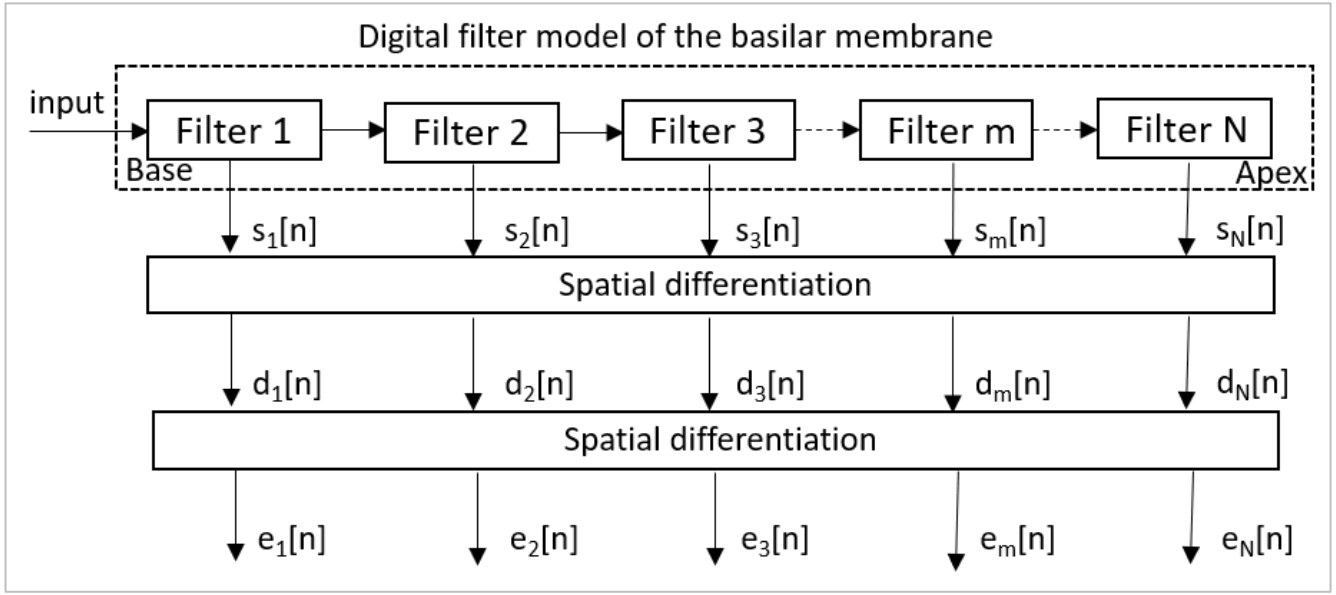
$$d_m[n] = s_m[n] - s_{m+1}[n]$$

For example: $d_1[n] = s_1[n] - s_2[n]$

- ✓ Applying spatial differencing a second time gives:

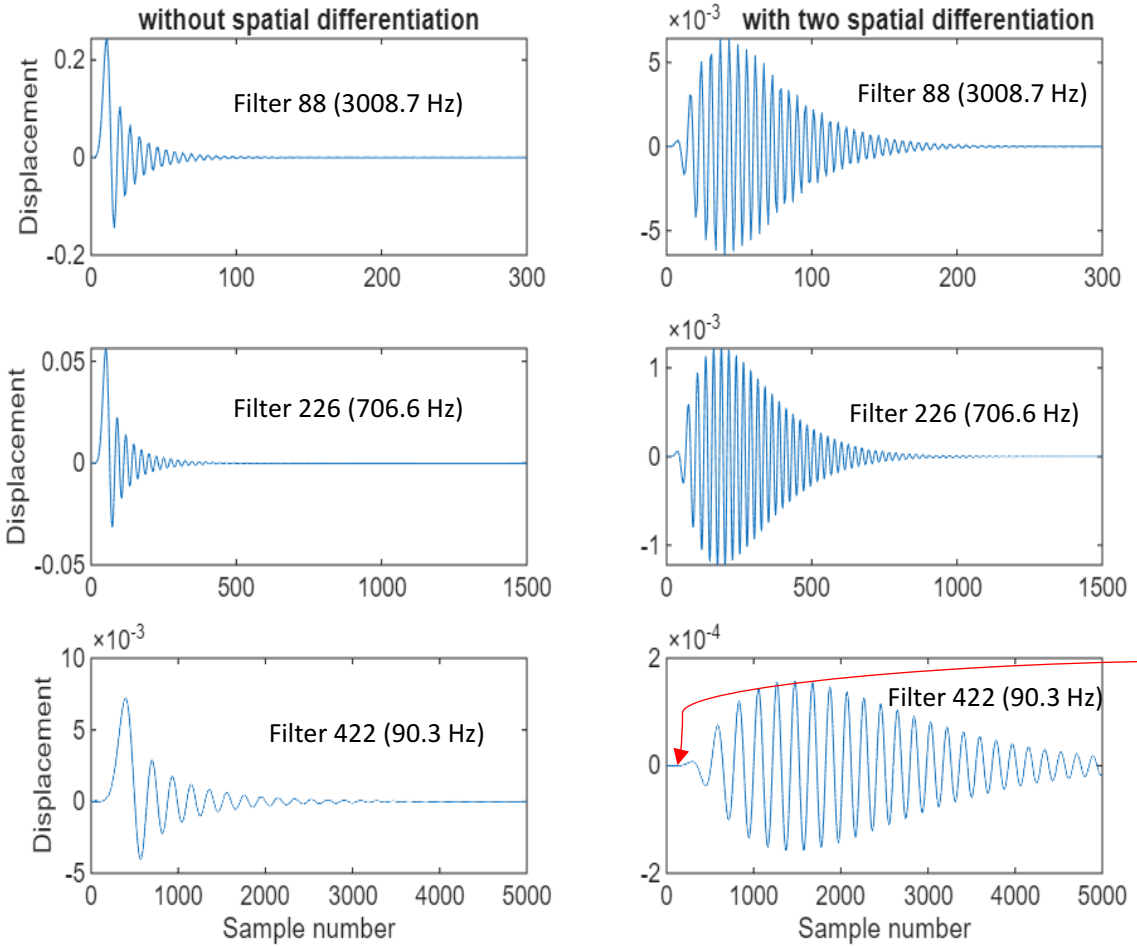
$$e_m[n] = d_m[n] - d_{m+1}[n]$$

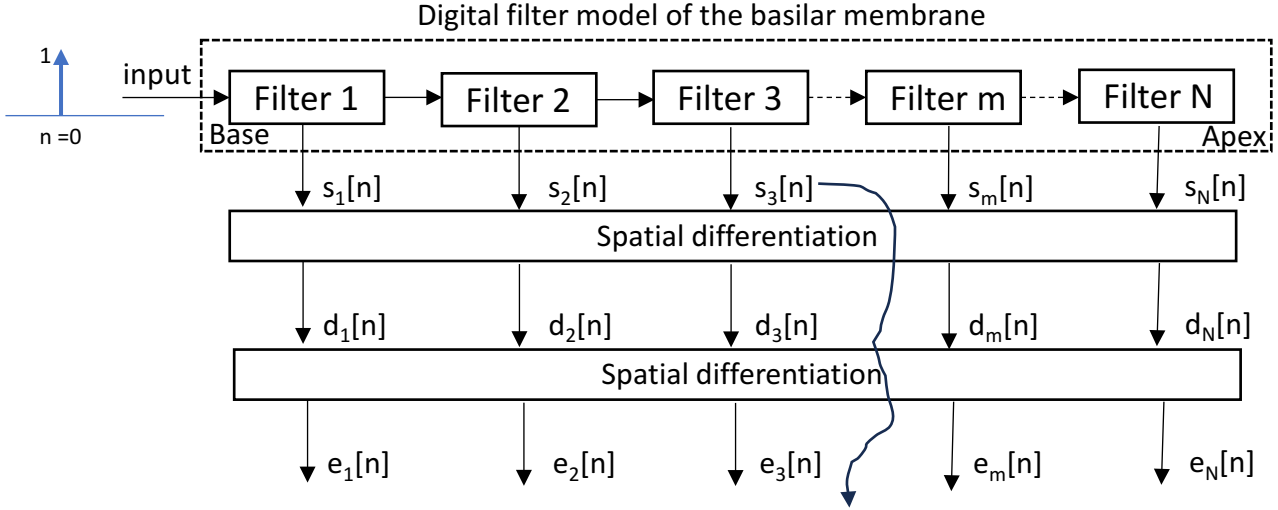
For example: $e_1[n] = d_1[n] - d_2[n]$



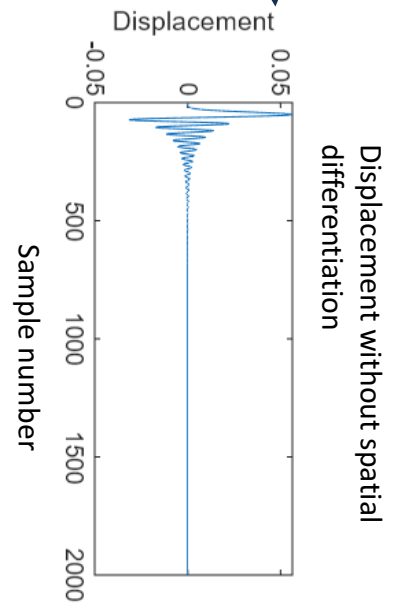
Are you on the right track?

✓ If your implementation is correct, the displacement impulse responses at the selected BM sections should resemble those shown below. Compare the responses before and after two stages of spatial differentiation





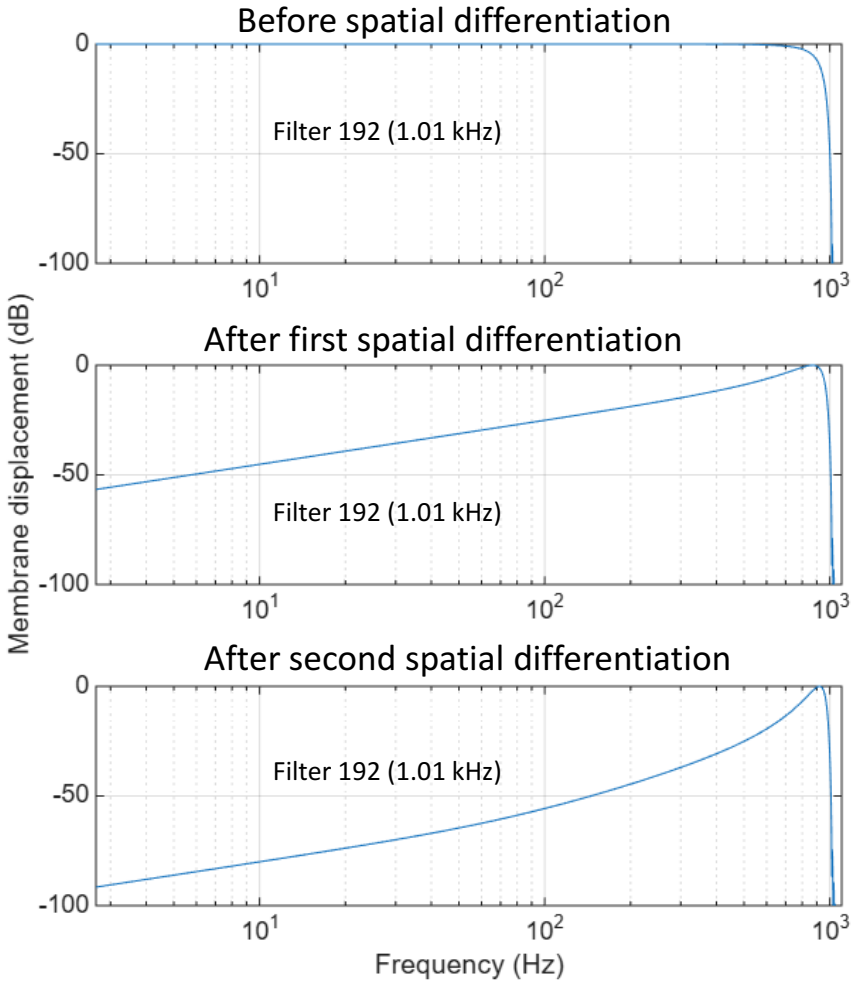
The travelling-wave delay is clearly visible in the filter responses.

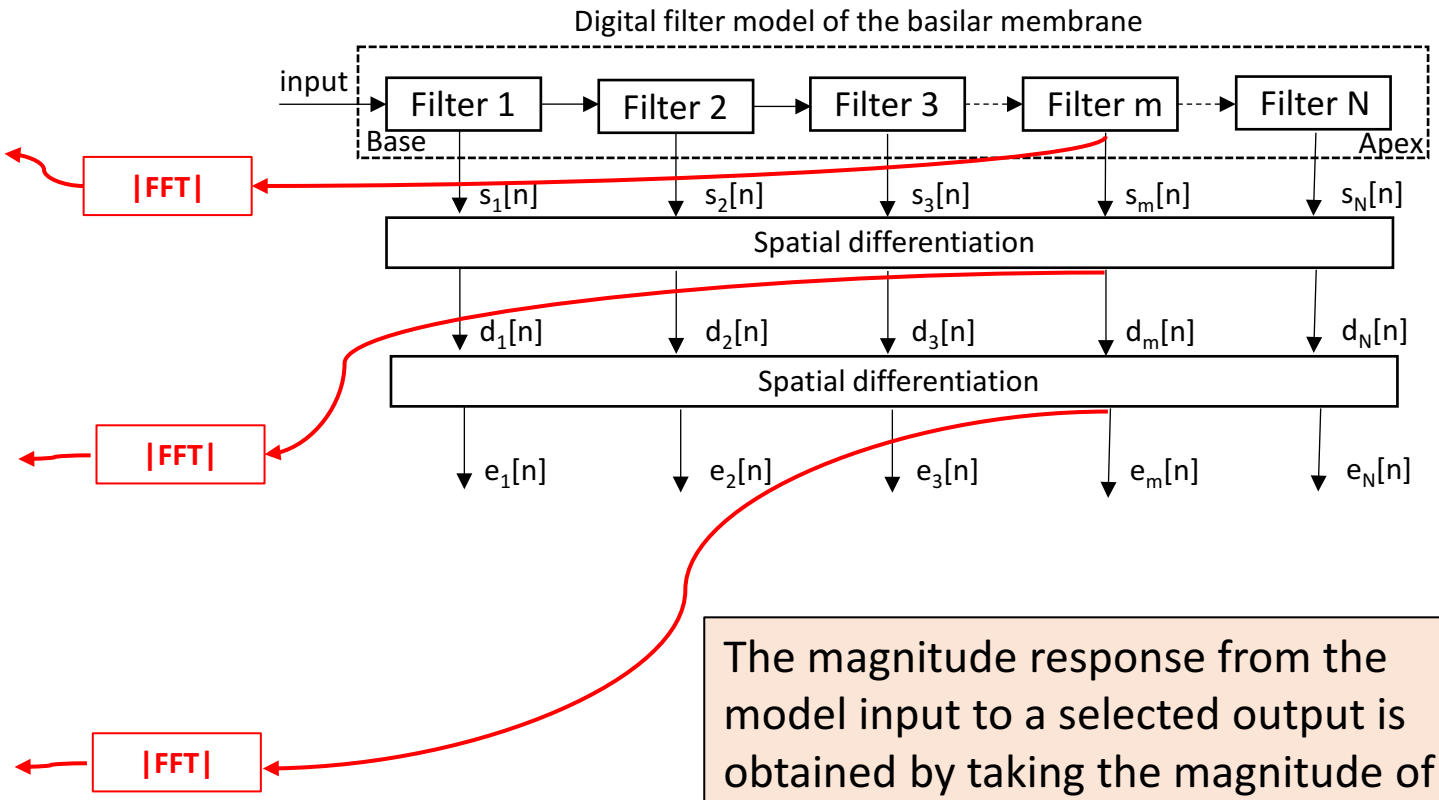


Sections towards the apex have lower oscillation frequencies and longer response durations. The travelling-wave delay becomes more apparent in the responses after spatial differentiation

Are you on the right track?

✓ If your implementation is correct, the magnitude responses at section 192 (approximately 1.01 kHz) should resemble those shown below, before and after each stage of spatial differentiation.





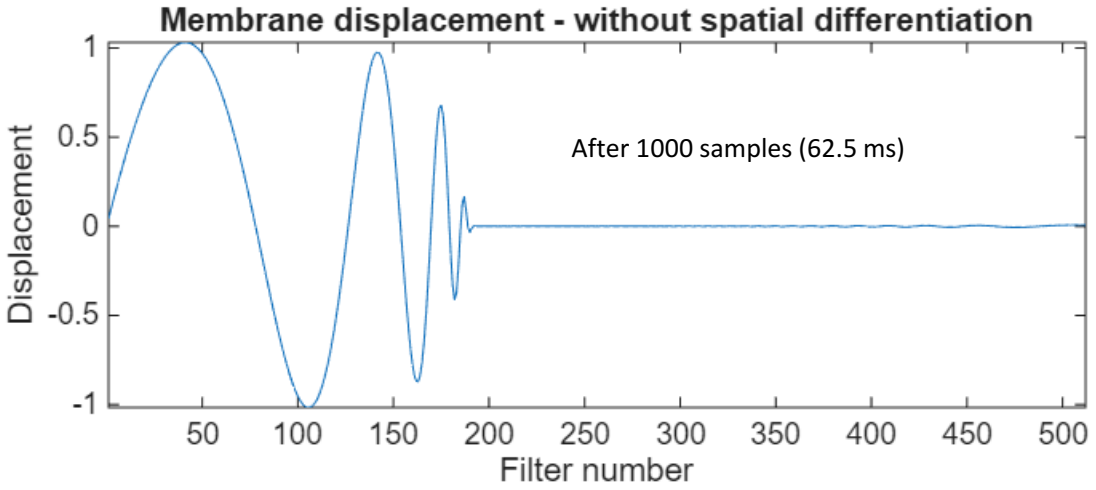
The magnitude response from the model input to a selected output is obtained by taking the magnitude of the FFT of its impulse response.

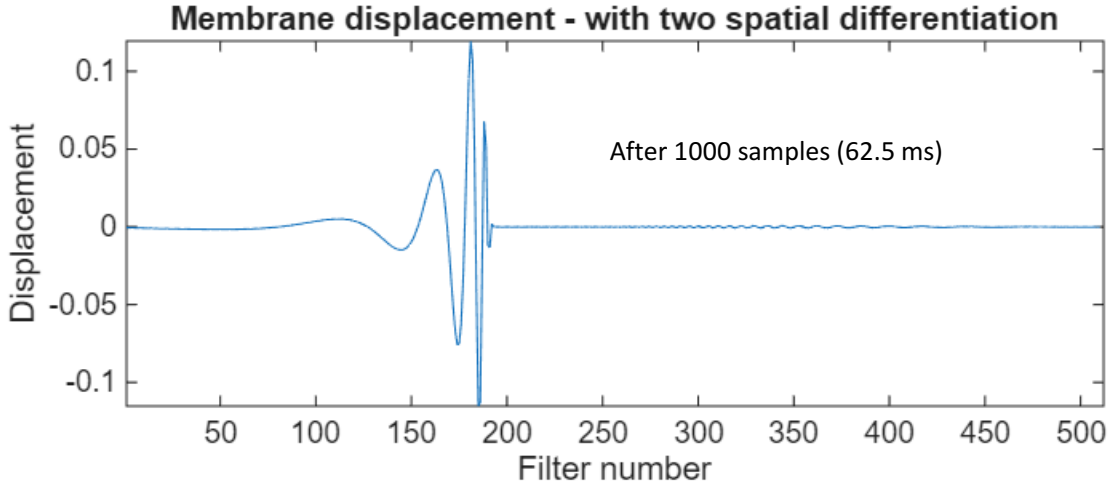
Magnitude responses at section 192 (1.01 kHz) of the 512-section cochlear model

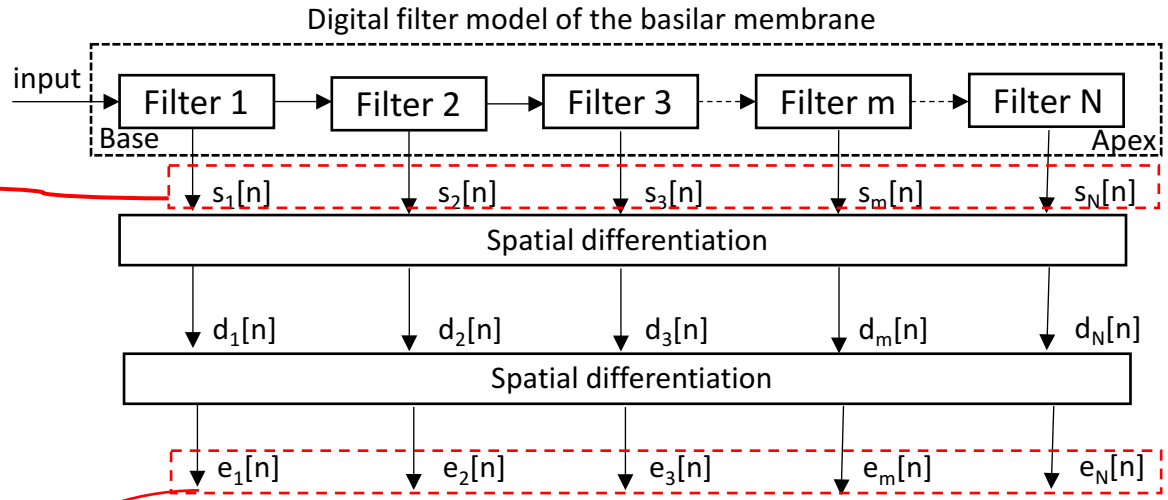
Two stages of spatial differentiation produce a more sharply tuned response, with greater attenuation at frequencies below the peak.

Are you on the right track?

- ✓ For a 1 kHz sinusoidal input, plot the displacement against section number at a selected time instant, before and after two stages of spatial differentiation. After spatial differentiation, the response is concentrated near the section tuned to the input frequency and decreases rapidly towards the apex beyond this region





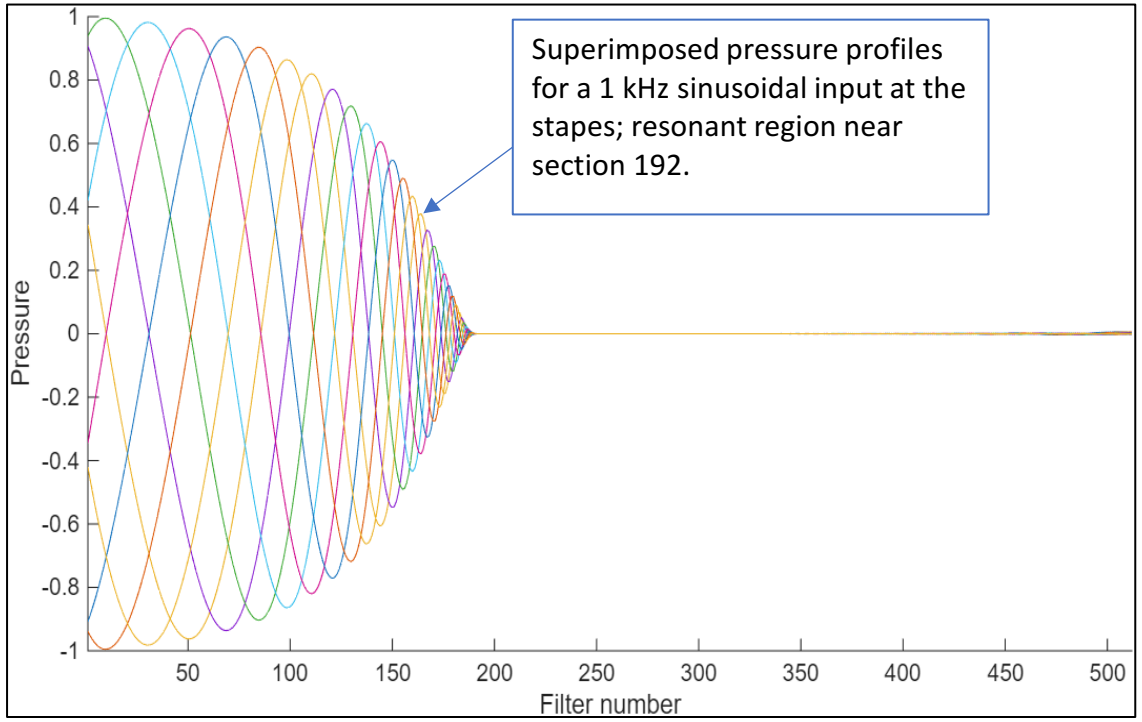


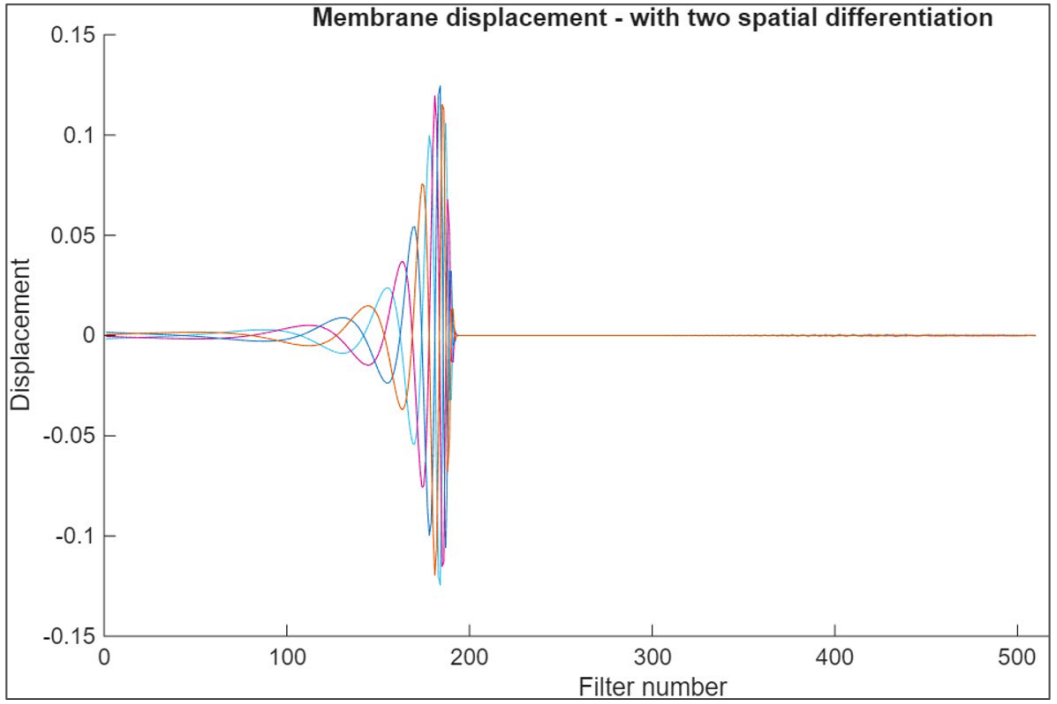
✓If your spatial differentiation is implemented correctly, the displacement pattern should resemble the lower plot for a 1 kHz sinusoidal input.

Are you on the right track?

- ✓ Plot pressure against section number at several equally spaced time instants after the initial transient has decayed. Superimpose these curves on the same axes to reveal the pressure envelope. For a 1 kHz sinusoidal input, the pressure decreases towards the resonant region near section 192 and becomes negligible beyond it.

- ✓ Plot the displacement after two stages of spatial differentiation against section number at several equally spaced time instants, after the initial transient has decayed. Superimpose these curves on the same axes to reveal the displacement envelope. For a 1 kHz sinusoidal input, the response is concentrated near the resonant region around section 192 and becomes negligible beyond it





Pass-Level Final Validation

1. Apply a sum of three equal-amplitude sinusoids, using one low, one middle and one high frequency of your choice. Generate 1000–3000 samples at f_s= 16 kHz ensuring all frequencies are below 7.5 kHz.

2. At the same selected time instant, plot displacement against section number in one figure using three subplots:
   - Before spatial differentiation.
   - After one stage of spatial differentiation.
   - After two stages of spatial differentiation

3. Select approximately ten sections spanning the frequency range of the cochlear model. Using their impulse responses, calculate and plot the magnitude responses from the model input to the selected displacement outputs. Present the results in two subplots: (a) Before spatial differentiation; (b) After two stages of spatial differentiation.

   Superimpose the selected responses in each subplot. Normalise each response to a peak of 0 dB and label the axes Frequency (Hz) and Normalised magnitude (dB). Comment on how spatial differentiation affects frequency selectivity.

Reflection

1. **Frequency analysis:** How does the model separate an input signal into different frequency components? Explain your results for an impulse, a single sinusoid and a sum of three sinusoids.
2. **Travelling-wave behaviour:** How do the position of maximum response and the response delay change with input frequency? Why does the pressure become negligible beyond the corresponding resonant region?
3. **Spatial differentiation:** How do one and two stages of spatial differentiation change the displacement patterns and magnitude responses? What evidence shows improved frequency selectivity?
4. **Quality factors:** How do changes in Q_p and Q_z affect bandwidth, frequency selectivity and the duration of the impulse response? Explain the trade-off between sharp tuning and rapid response.
5. **Role of the zeros:** Why are notch filters included between successive resonators? Predict and investigate what happens when the notch filters are removed or their frequencies are changed.
6. **A tone being removed:** If one component of a three-tone input is switched off, which sections are affected, and how quickly do their outputs decay? Relate your observations to propagation delay and filter ringing.
7. **Validation and learning:** What checks gave you confidence that your implementation was correct? Describe one difficulty or unexpected result, how you investigated it and what you learned.
8. **Model limitations:** Which aspects of cochlear behaviour are represented by this model, and which are omitted? Propose one improvement and explain how you would evaluate it.